NEWTON SCHOOL

PRE-CALCULUS | ANSWER KEY

Partial Fractions, Geometric Series & Telescoping

Student: Justin Koo **Date:** 2026-05-03

Standard: *CCSS.MATH.CONTENT.HSA-SSE.B.4 / HSA-APR.D.6.*

Part 1 — Algebra Foundations

Problem 1

Simplify $\dfrac{2/5 + 1/4}{1 - 3/10}$.

complex fraction | inside-out

Numerator: $\frac{2}{5} + \frac{1}{4} = \frac{8}{20} + \frac{5}{20} = \frac{13}{20}$.

Denominator: $1 - \frac{3}{10} = \frac{7}{10}$.

Divide: $\dfrac{13/20}{7/10} = \frac{13}{20} \cdot \frac{10}{7} = \frac{130}{140} = \frac{13}{14}$.

$$\boxed{\frac{13}{14}}$$

Problem 2 — Nested (3 layers)

Simplify $\dfrac{1}{1 - \dfrac{1}{1 + \frac{1}{3}}}$.

nested fraction | inside-out

Layer 1 (innermost): $1 + \frac{1}{3} = \frac{4}{3}$.

Layer 2: $\dfrac{1}{4/3} = \frac{3}{4}$.

Layer 3: $1 - \frac{3}{4} = \frac{1}{4}$.

Outer: $\dfrac{1}{1/4} = 4$.

$$\boxed{4}$$

Problem 3

$$\frac{1}{n(n+2)} = \frac{A}{n} + \frac{B}{n+2}.$$

partial fractions | gap 2 | used in Part 3 #4

Combine: $A(n+2) + Bn = 1 \Rightarrow (A+B)n + 2A = 1.$

Match: $A + B = 0, 2A = 1 \Rightarrow A = \frac{1}{2}, B = -\frac{1}{2}.$

$$\boxed{A = \tfrac{1}{2}, \quad B = -\tfrac{1}{2}}$$

Problem 4

$$\frac{1}{(3n-1)(3n+2)} = \frac{A}{3n-1} + \frac{B}{3n+2}.$$

partial fractions | used in Part 3 #2

Combine: $A(3n+2) + B(3n-1) = 1 \Rightarrow (3A+3B)n + (2A-B) = 1.$

Match: $3A + 3B = 0 \Rightarrow A = -B.$ Then $2A - B = 2A - (-A) = 3A = 1 \Rightarrow A = \frac{1}{3}, B = -\frac{1}{3}.$

$$\boxed{A = \tfrac{1}{3}, \quad B = -\tfrac{1}{3}}$$

Problem 5 — Factor first

$$\frac{1}{n^2+4n+3} = \frac{A}{?} + \frac{B}{?}.$$

factor + partial fractions | used in Part 3 #3

Factor denominator: $n^2 + 4n + 3 = (n+1)(n+3).$

Set $\dfrac{1}{(n+1)(n+3)} = \dfrac{A}{n+1} + \dfrac{B}{n+3}$. Combine: $A(n+3) + B(n+1) = 1.$

Match: $(A+B)n + (3A+B) = 1 \Rightarrow A + B = 0, 3A + B = 1 \Rightarrow 2A = 1 \Rightarrow A = \frac{1}{2}, B = -\frac{1}{2}.$

$$\boxed{A = \tfrac{1}{2}, \quad B = -\tfrac{1}{2} \quad \textbf{(over } (n+1) \textbf{ and } (n+3)\textbf{)}}$$

Part 2 — Geometric Series

Problem 1 — Finite

$$\sum_{k=0}^{5} 6\left(\tfrac{2}{3}\right)^k$$

finite formula | 6 terms

$g_0 = 6$, $r = 2/3$, upper power $n+1 = 6$.

$$S = 6 \cdot \frac{1-(2/3)^6}{1-2/3} = 6 \cdot \frac{1-64/729}{1/3} = 6 \cdot 3 \cdot \tfrac{665}{729} = \tfrac{11970}{729} = \tfrac{1330}{81}.$$

Decimal: ≈ 16.42.

$$\boxed{S = \frac{1330}{81} \approx 16.42}$$

Problem 2 — Infinite

$$\sum_{k=0}^{\infty} 6\left(\tfrac{2}{3}\right)^k, \text{ plus the gap with \#1.}$$

infinite formula | |r|<1 ✓

$|r| = 2/3 < 1 \Rightarrow$ converges. $S_\infty = \frac{6}{1-2/3} = \frac{6}{1/3} = 18$.

Gap with #1: $18 - \frac{1330}{81} = \frac{1458}{81} - \frac{1330}{81} = \frac{128}{81} \approx 1.58$. That's the tail $\sum_{k=6}^{\infty} 6(2/3)^k$.

$$\boxed{S_\infty = 18, \quad \textbf{gap} = \frac{128}{81} \approx 1.58}$$

Problem 3 — Shifted start index

$$\sum_{k=3}^{8} \left(\tfrac{1}{2}\right)^k$$

finite | index shift

Shift: let $j = k-3$, so $(1/2)^k = (1/2)^{j+3} = \frac{1}{8}(1/2)^j$. Sum becomes:

$$\sum_{j=0}^{5} \tfrac{1}{8}(1/2)^j = \tfrac{1}{8} \cdot \frac{1-(1/2)^6}{1-1/2} = \tfrac{1}{8} \cdot 2 \cdot \tfrac{63}{64} = \tfrac{63}{256}$$

Sanity (direct): $\frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \frac{1}{128} + \frac{1}{256} = \frac{32+16+8+4+2+1}{256} = \frac{63}{256}$ ✓

$$\boxed{\frac{63}{256}}$$

Problem 4 — Negative ratio with $g_0 \neq 1$

> $$\sum_{k=0}^{\infty} 8\left(-\tfrac{3}{4}\right)^k$$

infinite formula | sign care

$g_0 = 8$, $r = -3/4$, $|r| = 3/4 < 1$ ✓.

$$S_\infty = \frac{8}{1-(-3/4)} = \frac{8}{1+3/4} = \frac{8}{7/4} = \frac{32}{7}.$$

Sign of r doesn't affect convergence — only $|r|$ does.

$$\boxed{S_\infty = \frac{32}{7}}$$

Problem 5 — 3-digit repeating decimal

> Express $0.\overline{135} = 0.135135135\ldots$ as a fraction.

geometric series | repeat block = 3 digits

Chunk by 3 digits: $0.\overline{135} = 0.135 + 0.000135 + 0.000000135 + \cdots$

Geometric: $g_0 = \frac{135}{1000}$, $r = \frac{1}{1000}$.

$$S_\infty = \frac{135/1000}{1-1/1000} = \frac{135/1000}{999/1000} = \frac{135}{999}.$$

Reduce: $\gcd(135, 999) = 27$. So $\dfrac{135}{999} = \dfrac{5}{37}$.

Check: $5 \div 37 = 0.135135\ldots$ ✓

$$\boxed{0.\overline{135} = \frac{5}{37}}$$

Part 3 — Telescoping & Power Series IoC

Problem 1 — Telescoping (basic, gap-1)

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$$

telescoping | uses Part 1 #4: A=½, B=−½

From Part 1 #4: $\dfrac{1}{(2n-1)(2n+1)} = \frac{1}{2}\left(\dfrac{1}{2n-1} - \dfrac{1}{2n+1}\right)$.

$S_N = \frac{1}{2}\left[\left(\frac{1}{1} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{2N-1} - \frac{1}{2N+1}\right)\right] = \frac{1}{2}\left(1 - \frac{1}{2N+1}\right)$.

$N \to \infty$: $S_\infty = \frac{1}{2}(1-0) = \frac{1}{2}$.

$$\boxed{S_\infty = \frac{1}{2}}$$

Problem 2 — Telescoping (gap-1, new pattern)

$$\sum_{n=1}^{\infty} \frac{1}{(3n-1)(3n+2)}$$

telescoping | uses Part 1 #4: A=⅓, B=−⅓

From Part 1 #4: $\dfrac{1}{(3n-1)(3n+2)} = \frac{1}{3}\left(\dfrac{1}{3n-1} - \dfrac{1}{3n+2}\right)$.

Plug in $n = 1, 2, 3, \ldots$: denominators are $(2,5), (5,8), (8,11), (11,14), \ldots$ Note $(3 \cdot 1 - 1) = 2$ and $(3n+2)$ for n matches $(3(n+1)-1)$ for the next term — perfect telescoping.

$S_N = \frac{1}{3}\left[\left(\frac{1}{2} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{8}\right) + \cdots + \left(\frac{1}{3N-1} - \frac{1}{3N+2}\right)\right] = \frac{1}{3}\left(\frac{1}{2} - \frac{1}{3N+2}\right)$.

$N \to \infty$: $S_\infty = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.

$$\boxed{S_\infty = \frac{1}{6}}$$

Problem 3 — Telescoping (gap-2, factor first)

$$\sum_{n=1}^{\infty} \frac{1}{n^2+4n+3}$$

factor first | gap-2 telescoping | uses Part 1 #5

Factor: $n^2+4n+3=(n+1)(n+3)$. From Part 1 #5: $\dfrac{1}{(n+1)(n+3)} = \frac{1}{2}\left(\dfrac{1}{n+1} - \dfrac{1}{n+3}\right)$.

Write out the first several terms (denominators jump by 2):

$$S_N = \tfrac{1}{2}\left[\left(\tfrac{1}{2}-\tfrac{1}{4}\right)+\left(\tfrac{1}{3}-\tfrac{1}{5}\right)+\left(\tfrac{1}{4}-\tfrac{1}{6}\right)+\left(\tfrac{1}{5}-\tfrac{1}{7}\right)+\cdots+\left(\tfrac{1}{N+1}-\tfrac{1}{N+3}\right)\right]$$

The $-\frac{1}{4}$ kills the later $+\frac{1}{4}$, the $-\frac{1}{5}$ kills the later $+\frac{1}{5}$, etc. — but with gap 2, **two** terms survive at the start ($\frac{1}{2}$ and $\frac{1}{3}$) and two die out at the end:

$$S_N = \tfrac{1}{2}\left[\tfrac{1}{2}+\tfrac{1}{3}-\tfrac{1}{N+2}-\tfrac{1}{N+3}\right]$$

$N \to \infty$: $S_\infty = \frac{1}{2}\left(\frac{1}{2}+\frac{1}{3}\right) = \frac{1}{2}\cdot\frac{5}{6} = \frac{5}{12}$.

$$\boxed{S_\infty = \frac{5}{12}}$$

Problem 4 — Telescoping (gap-2)

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$$

gap-2 telescoping | uses Part 1 #3

From Part 1 #3: $\dfrac{1}{n(n+2)} = \frac{1}{2}\left(\dfrac{1}{n} - \dfrac{1}{n+2}\right)$.

Write out (gap 2 again):

$$S_N = \tfrac{1}{2}\left[\left(1-\tfrac{1}{3}\right)+\left(\tfrac{1}{2}-\tfrac{1}{4}\right)+\left(\tfrac{1}{3}-\tfrac{1}{5}\right)+\left(\tfrac{1}{4}-\tfrac{1}{6}\right)+\cdots+\left(\tfrac{1}{N}-\tfrac{1}{N+2}\right)\right]$$

Survivors: first two (1 and $\frac{1}{2}$), and the last two negative terms vanish.

$$S_N = \tfrac{1}{2}\left[1+\tfrac{1}{2}-\tfrac{1}{N+1}-\tfrac{1}{N+2}\right]$$

$N \to \infty$: $S_\infty = \frac{1}{2}\left(1+\frac{1}{2}\right) = \frac{1}{2}\cdot\frac{3}{2} = \frac{3}{4}$.

$$\boxed{S_\infty = \frac{3}{4}}$$

Problem 5 — FINAL CHALLENGE: Power Series IoC

> Find IoC and $P(x)$ for $\sum_{k=0}^{\infty} \left(\frac{x}{2}\right)^k$.

power series | IoC | complex fraction (Part 1)

Step 1 — Identify g_0 and $r(x)$. At $k = 0$: $g_0 = (x/2)^0 = 1$. Ratio $r(x) = \frac{x}{2}$.

Step 2 — Convergence.

$$\left|\frac{x}{2}\right| < 1 \implies \frac{|x|}{2} < 1 \implies |x| < 2 \implies -2 < x < 2$$

IoC $= (-2, 2)$.

Step 3 — Sum function (clean up the complex fraction).

$$P(x) = \frac{g_0}{1 - r(x)} = \frac{1}{1 - x/2}$$

Multiply top and bottom by 2 (Part-1 trick):

$$P(x) = \frac{1 \cdot 2}{(1 - x/2) \cdot 2} = \frac{2}{2 - x}$$

Sanity checks:

- $x = 0$: series $= 1$, formula $= \frac{2}{2} = 1$ ✓.
- $x = 1$: series $= 1 + \frac{1}{2} + \frac{1}{4} + \cdots = 2$, formula $= \frac{2}{2-1} = 2$ ✓.

$$\boxed{\textbf{IoC} = (-2, 2), \quad P(x) = \frac{2}{2 - x}}$$